Classifications

Linear: $\alpha_1 x_1(t) + \alpha_2 x_2(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t) \ \forall \ \alpha_1, \alpha_2$
To prove, sum first, then transform, and prove equal to, transform first then sum.
Energy:

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt = \int_{t_1}^{t_2} (x_{Re}^2(t) + x_{Im}^2(t)) dt$$

for total energy, $t_1 = -\infty, t_2 = \infty$.
Power:

$$P_{avg} = \frac{1}{t_2 - t_1} E = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} (x_{Re}^2(t) + x_{Im}^2(t)) dt$$

(In discrete time, the denominator is $n_2 - n_1 + 1$). For total, both limits tend to ∞.
Period: $\exists T \neq 0 | x(t) = x(t+T) \ \forall t \implies$ periodic.
Minimum T that fulfills conditions is T_0, fundamental period. For two added signals, $T = LCM(T_1, T_2)$.
$T(\sin(\omega t)) = \frac{2\pi}{\omega}$
Parity:
Even (+): $x(t) = x(-t)$
Odd (-): $x(t) = -x(-t)$

$$x = even + odd = \frac{x(t) + x(-t)}{2} + \frac{x(t) - x(-t)}{2}$$

Complex Exponential:

$$\begin{aligned} x(t) = Ce^{\alpha t} &= |C| \cdot e^{\sigma t} \cdot e^{j(\omega t + \phi)} \\ &= |C| \cdot e^{\sigma t} \cdot (\cos(\omega t + \phi) + \sin(\omega t + \phi)) \end{aligned}$$

$\omega = 2\pi / T$ is the fundamental frequency.

$$P(t) = |C|^2 w^{2\sigma t}$$

$$E_\infty = \infty, P_\infty = C^2$$

HRCEs: $x_k(t) = e^{jk\omega_0 t}$

Transformations

Time Shift:$x(t) \to x(t - t_0)$.Moves signal right by t_0.
Time Reversal:$x(t) \to x(-t)$.Reflects signal about y-axis.
Time Scale:$x(t) \to x(\alpha t)$.Widens signal by $\frac{1}{\alpha}$.

Complex Exponential

DTCE: $x[n] = |C| \cdot e^{\sigma n} \cdot e^{j\omega n + \phi}$
Has exponential envelope $|C|e^{\sigma t}$.

$$N_0 = LCM\left(T = \frac{2\pi}{\omega}, 1\right)$$

For sum, $N = LCM(N_1, N_2)$.
If no π in ω, or ω is irrational, the signal is not periodic.
$0 \leq \omega < 2\pi$. Speed up for $\omega \approx \pm\pi$, slow at borders.
DTHRCEs: $x_k[n] = e^{jk(\frac{2\pi}{N})n}$
Only N distinct members in the HRCE: $k = 0 \dots N-1$.

Special Signals

DT:
$\delta[n] = 0 \ \forall \ n \neq 0, \ \text{else } 1.$
$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = 1 \ \forall \ n \geq 0$
$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$
CT:
$\int_{-\infty}^{\infty} \delta(t) = 1$

System Classifications

Series:$x \to [S1] \to [S2] \to y$
Parallel:

$$\begin{array}{ll} & \uparrow \to [S1] \to \downarrow \\ x & \qquad + \to y \\ & \downarrow \to [S2] \to \uparrow \end{array}$$

Feedback:

$$\begin{array}{l} x \to + \to [S1] \to y \\ \qquad \uparrow \leftarrow [S2] \leftarrow \downarrow \end{array}$$

Properties

Memory: if $y(t) = f(x(T)) | T \neq t_0$, the system has memory.
Invertible: $x \to [S] \to [S^{-1}] \to x$

$$\forall x_1, x_2 | x_1 \neq x_2 \implies y_1 \neq y_2$$

Can be proven if x as a function of y is defined across all t.
Causal: $y(t) = \sum_{T=-\infty}^{t} f(x(T))$. No dependence on future values of t.
Stable: $|x(t)| < B \implies |y(t)| < M$ for finite B, M.
Time Invariant: $x(t) \to y(t) \implies x(t - t_0) \to y(t - t_0)$. Scaling, or having a function of t outside, like $(tx(t))$, will be time invariant.

LTI

Linear and Time Invariant system.